Computers in Human Behavior 56 (2016) 83–92

Contents lists available at ScienceDirect

Computers in Human Behavior

journal homepage: www.elsevier.com/locate/comphumbeh

Exploring college students' generational differences in Facebook usage

Svetlana Čičević [a], Andreja Samčović [a,*], Milkica Nešić [b]

[a] Faculty of Transport and Traffic Engineering, University of Belgrade, Serbia
[b] Faculty of Medicine, University of Niš, Serbia

ARTICLE INFO

Article history:
Received 19 February 2015
Received in revised form
9 November 2015
Accepted 18 November 2015
Available online xxx

Keywords:
Facebook
Social networking sites
Internet
Education
Communication technology

ABSTRACT

In recent years the usage of social networks among students has become the subject of numerous studies. In this study we were interested in differences in patterns of online behaviour and the use of Facebook (FB) among the second and third generation of students. The results showed that 66% of the second year students had a FB account, while in the third generation all of the respondents had a FB account. One third of the second year students avoid any involvement in FB. Lack of interest was one of the main reasons the respondents indicated for not being engaged in FB, as well as their unwillingness to put personal information online. Currently, most of them are satisfied using FB, and do not plan to turn off their profile. The newer generations of students use FB mainly for overcoming boredom. Based on the results of the study, it can be concluded that the time spent online increases, as well as the number of FB users, and also the number of virtual friends increases in order to exchange information.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Today, we can hardly imagine our lives without the social networks as they are the most widespread type of communication. Nowadays many social network sites (SNS) exist, offering a huge variety of services targeted at diverse audiences all over the world. A vast number of people sharing similar target interests, viewpoints and goals are connected and unified thanks to online social network sites. According to Boyd and Ellison (2007), an online social network site can be defined as a set of web-based services that allow individuals to: (1) construct a public or semi-public profile within a bounded system, (2) articulate a list of other users with whom they share a connection, (3) view and transverse their list of connections and those made by others within their system. Since most of SNS reflect real-life social relationships very faithfully, it seems that their primary role is to articulate existing social networks, rather than to extend current or to produce new kinds of networks.

When users join a social network, they start by creating a profile, and connecting with existing friends, as well as meeting various other needs of communication. Members use such sites for multiple purposes. While the initial motivation was to maintain communication and interpersonal relationships, additional popular activities include updating various activities, photo sharing and archiving events, getting news about their friends' future activities and/or upcoming major events. Moreover, users can add friends, send messages, and add new information in their profiles to notify friends about new developments.

The motivation behind the use of such networks and interaction patterns in the context of social media are of significant research interest because they allow us to investigate digital traces of human activities. For instance, a recent study reveals different behaviour patterns in reply interactions between friends and non-friends, with reply interactions more likely to occur between non-friends (Cattuto & Hotho, 2011).

In addition, while younger generations use social networks to maintain contacts with many people, regardless of physical distance, one has to recognize that SNS have drawbacks. One of the main challenges of today's information society is the problem of data security that users leave on social networking sites (Samčović & Čičević, 2013).

When it comes to social networking, the first association is linked to Facebook (FB), the largest and most popular virtual community of its kind in the world. These new forms of social networking impact on Facebook as a social networking site of particular interest to researchers due to its heavy usage patterns and its technological capabilities, which allow the bridging of online and offline relationships (Boyd & Ellison, 2007).

Another advantage of social networking is that it is now much

* Corresponding author.
E-mail addresses: s.cicevic@sf.bg.ac.rs (S. Čičević), andrej@sf.bg.ac.rs (A. Samčović), milkica@medfak.ni.ac.rs (M. Nešić).

http://dx.doi.org/10.1016/j.chb.2015.11.034
0747-5632/© 2015 Elsevier Ltd. All rights reserved.

easier to solve various problems, like the so-called "law of weak ties" (Teng, 2015). Weak ties are the links to people who are not friends but acquaintances, met at work or through real friends. Many users have mostly "weak ties" among FB friends. This provides advantages, if we find ourselves in a situation where we need an answer to a problem, since it is more likely that someone among our acquaintances, would be able to offer a solution. It is possible that our close friends may not know the solution, no matter how hard they try, because they move in similar circles as we, receive similar information and know our habits.

Today, social networks, as well as the internet and mobile phones have become universal media and something without which we cannot act optimally. This is particularly relevant for young people who prefer to express their thoughts and opinions on various subjects using short comments (Kim, Hwang, & Choi, 2013).

To date, schools and institutions are experiencing many technological advancements and changes in techniques in improving their teaching methods to meet the growing demands of the young generation of students who are very fond of using social media as often as possible. Institutional practice will be both driven and shaped by social media in the near future. There have been various overviews and opinions which recognized major advantages of social media use in higher education, which comprise enhancing relationship, improving learning motivation, offering personalized course material, active participation, critical thinking and developing collaborative abilities (Dawson, 2008).

Learning may occur as part of education, personal development, schooling or training. There are many styles of learning, which can be adopted to make students comfortable while learning. Nowadays, technology is a boon to students as it helps them to learn in the way they prefer. Social networks can also be used in order to enhance learning for young people with attention difficulties in various settings such as academia, education or entertainment (Drigas, Ioannidou, Kokkalia, & Lytras, 2014).

Most of the social network users are young individuals, and indeed the majority of them are university students (Mazman & Usluel, 2011). The relationship between youth and their involvement in social network sites, especially FB, has attracted much research that focused on young people's online activities (Mazman & Usluel, 2011). Besides the usual analysis which has been carried out for each of the social networking sites and comparison between them, it is also useful to provide appropriate investigation of the usage patterns of various categories of social networks users. Thus, the need arises to assess user experience, acceptance reasons for use and transitions from one network they currently use to other networks, or to a completely new kind of networks. The aforementioned reasons, along with everyday teaching practice and the perceived promising importance of one of the most popular social media, Facebook, for knowledge sharing and involvement in educational settings, motivated us to explore and better understand, if there are any generational differences among college students in terms of Facebook usage.

The rest of the paper is structured as follows. The next section describes the beginnings of Facebook use in modern world. The third section reviews prior studies of the use of Facebook in education. The fourth section presents an attempt to address specific issues related to generational differences in Facebook usage on the basis of students' answers to the questionnaire. The fifth section presents the findings of our study. Finally, we conclude the contribution by describing the implications for both research and practice, as well as the limitations of the current study and future research directions.

2. Facebook use in modern world

The last few years have witnessed an increased interest in examining the use of Facebook, one of the most popular – and misunderstood – social networking sites on the internet. The widespread use of Facebook by individuals around the world attests to its potentiality to play important social, educational, and several other communicative and interactive roles. So far, most Facebook researchers have employed lean measures such as minutes per day spent using the service and/or the likelihood of engaging in (actual use of) a social activity using Facebook (Ellison, Steinfield, & Lampe, 2007).

According to 2014 Facebook statistics, the total number of monthly active Facebook users worldwide is over 1,310,000,000; whereas the total number of mobile FB users is 680,000,000; among different education levels and social classes, including a variety of companies and universities (Facebook.com, 2014). In Serbia, the percentage of monthly active Facebook users is 48%, while the percentage of mobile FB users reaches 8% (Marketingmreza.rs, 2014).

The number of Facebook users in Central and Eastern Europe (including Russia) was 122 million in 2014, and rise to 166.4 million by 2017 (with Russian users making up 49 million of that total). The typical (median) teen Facebook user has 300 friends, while the typical teen Twitter user has 79 followers (Madden et al., 2013).

According to the Statistical Office of the Republic of Serbia, more than 2.4 million people use the internet every day or almost every day. About 55.8% of households have an internet connection, while the European average of the internet penetration is 50%–70% (Webrzs.stat.gov.rs, 2014). About 41% of people in Serbia have never used the internet and do not intend to, because they feel they do not need it.

Serbia, with as many as 3,285,000 registered accounts on the Facebook, is one of the leading countries in Europe, according to the proportion of the overall population, as well as 47th in the world, taking into account the total number of accounts. Facebook users also represent 85% of the total number of internet users in Serbia. The data show that FB in Serbia is more used by males (56%) than females (44%), and most of them are between 18 and 34 years (61%). However, surprisingly, the highest growth rate of users is in the age group of over 65 years (Webrzs.stat.gov.rs, 2014).

The data, collected by the EDUCAUSE Center for Applied Research (ECAR) from a sample of 36,950 students from 126 U.S. universities and one Canadian university, showed that of the 90% of students who use social networking websites, 97% of which said they used Facebook (Junco, 2012a). This 97% of users reported actively engaging on the site daily (Salaway et al., 2007). In a more recent study, Ellison, Steinfield, and Lampe (2011) extended their previous work and found that engaging in social information-seeking behaviours on Facebook (for instance, to learn more about people with whom the user has an offline connection) was related to increased social capital, while using Facebook to maintain close ties and to meet strangers without any previous offline connection was not. Because of the viral speed with which the use of Facebook and other social networking technologies is growing among college-aged youth, there is a great deal of popular and academic interest in the impact that these technologies may have on student academic outcomes. The most popular and common online activities of students are visiting the websites of social networks (Jones & Fox, 2009; Lenhart, 2008).

3. The use of Facebook in education

As one of the most popular social network, Facebook is a tool for communication, very similar to e-mail, as it is essentially based on

the same technology. It is expected that in the future, an increasing number of higher education institutions will use this kind of communication. However, at present time, most of the online communication between teachers and students is carried out by e-mail. Teachers will certainly be willing to adopt technology which is proved to facilitate communication with students. Even more significant is the social aspect of FB. In teaching experience, it looks promising to establish appropriate relationships with students using FB and similar technologies. While many teachers have FB pages and are actively connected with their students, some experiences suggest that there must be awareness of the problems that such activities could carry (Flynn, 2009; Young, 2009).

However, with the growth in the adoption of social networking tools such as Facebook and Twitter in teaching and learning, there is a need to investigate the reasons for the adoption of the tool(s), the benefits that students perceive, and the challenges they face. The empirically grounded findings of students' perceptions and experiences will help to inform institutions, policy makers and educators who are planning to introduce social software tools in their courses and institutions (Shailey, 2009).

As more students are now spending a significant extent of their time participating in online social network sites, they could become a valuable tool to enhance student learning. As the site capabilities expanded beyond networking, individuals have begun to use Facebook for a variety of non-networking related reasons (Hayes, van Stolk-Cooke, & Muench, 2015). Facebook has become a big part of students' lives and that is the reason why Facebook can have a major impact on education.

Manca and Ranieri (2013) identified five main educational uses of Facebook: 1) to support class discussions and helping students engage in collaborative learning; 2) developing content; 3) sharing educational resources; 4) delivering content to expose students to extra-curricular resources; and 5) to support self-managed learning.

Facebook is considered to be an educational tool, since it enables peer feedback, interaction, and learning in a social context. Students generally expressed the opinion that Facebook provides a simple environment for them to upload and edit their ideas. They found Facebook as easy, convenient, user-friendly and simple for academic discussion. Discussing a topic in a group can lead to forming teams based on students with similar interests (Ku, Tseng, & Akarasriworn, 2013). When working in teams, the group's cohesiveness increases which results in a higher level of efficiency and understanding among students. Team-based learning can enhance education in multiple ways. This is especially important since possessing excellent teamwork skills is one of the important factors for employers in the job market. This is another reason why social media have an increasingly significant impact on human resource practises and the potential to be used in employment decisions, keeping in mind the, certainly, ambiguous effects they may have on future law and policy development (Drouina, O'Connor, Schmidt, & Miller, 2015).

Universities that use technology can positively motivate students in their course work. An increase in student motivation can often result in a positive impact on student engagement, grades, and on personal psychological aspects. Using Facebook for academic collaboration can enhance college freshmen's bonding social capital, which facilitates social adjustment to college (Gray, Vitak, Easton, & Ellison, 2013). Understanding the students' learning experiences and changes of their learning patterns is essential for the design of the instruction methods when aiming to maximize students' learning effectiveness. Creating such learning environment will not be possible without understanding differences among students (Chen, 2011).

Mazer et al., (2007) found that FB is starting to be used by both, students and teachers. There is an increasing number of users, who identify themselves as teachers. They suggest that teachers perceived the students' usage of FB as a constructive way to foster positive relationships with their students. FB could have a positive impact on students' learning outcomes, and the teachers' engagement on FB demonstrates their acceptance and understanding of contemporary student culture. However, the accompanying consequence could be the modification of their academic credibility. This interaction among teachers and students plays a key role in improving the quality of online courses (Roblyer et al., 2010). Researchers in the social sciences are trying to study the key features of FB use among younger generations, to understand their online interaction, communication and identity as members of online communities (Dong, 2008).

An increasing number of publications have been dealing with the exploration of the way that young people, especially students use information and communication technology (Cotten, 2008; Jones & Fox, 2009; Pasek & Hargittai, 2009; Roberts & Foehr, 2008). Although social networks were designed primarily for communication purposes, they indicate the transition to other areas of the youth life, including education. Karlin (2007) reports that almost 60% of students use social networks in order to communicate on issues related to education and more than 50% use social networks for the exchange of experience doing their homework. Zhang, de Pablos and Zhang (2012) provided some theoretical contributions on e-learning and knowledge management, and also discussed the use of technology in online engineering education system in the context of social networks. Social networks can facilitate students in learning fundamental programming concepts, or for instance in creating animated movies and simple video games (Xi, Liu, & de Pablos, 2014).

Nagel and Kotze (2010), suggested that it is necessary to pay attention to the activity on social networks and interpersonal relations as a new form of opportunities for learning, training and exchange of experience and cooperation. For example, joining groups where users share similar interests has pedagogical potential that can be used in constructive ways (Blattner & Fiori, 2009). Chang and Heo (2014) suggested the potential of relationships, time spent on Facebook, number of Facebook friends, and gender difference as frameworks to explain self-disclosure behaviour on FB. Park, Lee, and Kim (2012) discovered that various dimensions of Facebook-specific network characteristics were positively associated with usage patterns, including time spent on FB and posting messages and photos. In another study by Baek, Holton, Harp, and Yaschur (2011), user motivations for engaging in link-sharing through Facebook are explored, including motivations that add to current viewpoints of social network sites as information hubs.

3.1. The current study

The introduction of Facebook in the process of education is relatively new and there is little research on their acceptance. We believe that Facebook is appropriate for the current study as it is the most popular online social networking site among university students. This study will provide preliminary data on the change of habits in using FB by different generations of students. Our survey will attempt to provide answers on the following questions:

- Why do students use FB?
- What percentage of students who use FB for daily activities and social interactions, and how much for academic purposes?
- How long do students have a FB account, how often do they visit for and how much time they do spend on FB?
- What is their average number of friends on FB?

4. Methods

As an integral part of the educational process reforms at the University of Belgrade – Faculty of Transport and Traffic Engineering, a number of initiatives have taken place to incorporate pedagogical innovations in university courses (Pejčić-Tarle, Čičević, Davidović, Petrović, & Bojković, 2007). The main objective was, to improve the educational process, through engaging in receiving feedback and analysing the students' responses and attitudes. After a period of years, we observed that there are behavioural, as well as academic performances differences, among students from various courses. By interviewing our students, we found that many of them attributed their learning difficulties to the form in which course material was presented. Some students found they had difficulties learning in situations where the course material was only presented orally, while others reported similar difficulties when the material was primarily in written form. Still other students experienced difficulties with ideas that were presented in graphics or without any associated practical experiences. These insights prompted us to focus on sensory modalities as a learning style dimension that had some pre-eminence over others (Čičević, Nešić, & Samčović 2012).

Auditory and tactile learning style preferences were found to be dominant among our students. Near 40% of students prefer individual work, and 24% indicates that they may have difficulties learning in group. Since this study commenced, we continued the analysis of students' attitudes and behaviours, in each semester, either through interviews or questionnaires, and found that changes in learning styles occurred during the course of their academic life.

Peters, Gareth, and Peters (2005) also found significant differences in the preferred learning styles between the first and third years of the student population for the auditory learning style, with the third year group demonstrating significantly higher preference for the auditory category. This appears to be congruent with the significantly higher level of autonomy in the third year, as the definition of the auditory learning style included the ability to learn from class discussions and conversing with tutors (Pejčić-Tarle et al., 2007).

In the early beginnings of the introduction of the e-learning concept, our students appeared to favour traditional, passive and linear forms of learning and teaching. Indeed, their expectations of integration of digital technologies in teaching focused around the use of established tools within conventional pedagogies. They expect technology-enhanced learning methods to reflect conventional learning and were uncomfortable with the application of social technologies in the educational contexts. Over the following years their attitudes started to gradually change towards adopting new forms of learning (e.g. blended learning, for more details see Quan-Haase, 2007), as well as considering the use of social networking tools, in the instructional process (Čičević, Samčović, & Čubranić-Dobrodolac, 2012).

The introduction of online social network sites opens up opportunities for today's learners. In order to provide effective learning in a variety of contexts, whether it takes place in traditional classrooms, online learning, or blended learning that integrates classroom-based lessons with online discussions, it is necessary to consider factors affecting student's learning before incorporating online social networking in education (e.g. socio-economic backgrounds or life circumstances). Recognizing the realities of students' experience of learning will provide a better understanding of how to evolve teaching practises. One of the current challenges in teaching, particularly in the case of distance teaching over media such as the internet, is to meet the needs of students from a heterogeneous population.

Approaches to learning are dynamic and change over the course of a student's period of study. Given the fast-paced nature of changes in this domain, it is natural that students in the digital age, submerged in the advances of information and communication technologies (ICT), have particular characteristics and different learning habits that lead them to use technology differently in comparison to digital immigrants (i.e. people born prior to 1980) as defined by Prensky (2001). But, even among those millennial students marked by their exposure to digital technologies from an early age, a variety of subgroups exists, in terms of their use of social media. This is indeed the research question we seek to examine in this study.

4.1. Participants

In this study we were interested in understanding the differences in the patterns of online behaviour and the use of Facebook among two generations of students at the University of Belgrade – Faculty of Transport and Traffic Engineering. The sample consisted of 200 participants, 100 students in the second year of the study (60% males, 40% females) and the same number of third year students (45% males and 55% females). The subjects were students in an introductory psychology course and voluntarily agreed to participate in the study.

Students completed a survey consisting of multiple-choice questions, which, in addition to demographic information, included information about their previous experience in using the internet, the frequency and temporal aspects of Facebook usage, the type of activity on FB, number of friends, motivation for FB connectedness and the type of activities they engaged.

In this study, an exploratory approach was adopted aimed at providing descriptive information about the use of Facebook by college students, with a focus on generational differences in patterns of Facebook usage. Also, Chi-square tests were performed and evaluated to test differences between the two generations of university students.

5. Results of the research

The results showed that all third-year students have a Facebook profile as opposed to 66% of the second year students. A third of second year students avoided getting involved with Facebook. The main reasons why they did not join FB, as was specified, were: that they were simply not interested in FB, or did not want their personal information to be available on the internet.

Furthermore, most of their friends did not have a FB account. In addition to lack of time, they felt that using Facebook was a fashion trend that they did not have an interest to follow.

Currently, most of the subjects are satisfied with Facebook and have no intentions of removing or deleting their profile. One of the main reasons why third year students use Facebook is to overcome boredom. Moreover, second year students showed better academic achievement (χ^2 (3) = 25.05, p < .0001, Cramer's V = .35).

The two generations of university students differ significantly with respect to their experience of internet use (χ^2 (3) = 31.71, p < .0001, Cramer's V = .4). The majority of students in their second year of the studies have more experience in internet use, as measured by the number of years (3 or more) spent, as opposed to the third year students, where half of the respondents have experience of 2–3 years, as shown in Fig. 1.

When considering online gaming, 74% of the second year students were not interested to play. Chi-square test with Yates' continuity correction revealed that a significantly higher percentage of the third year students, around 70%, play online games (χ^2 (1) = 37.04, p = .000) in comparison to the younger generation

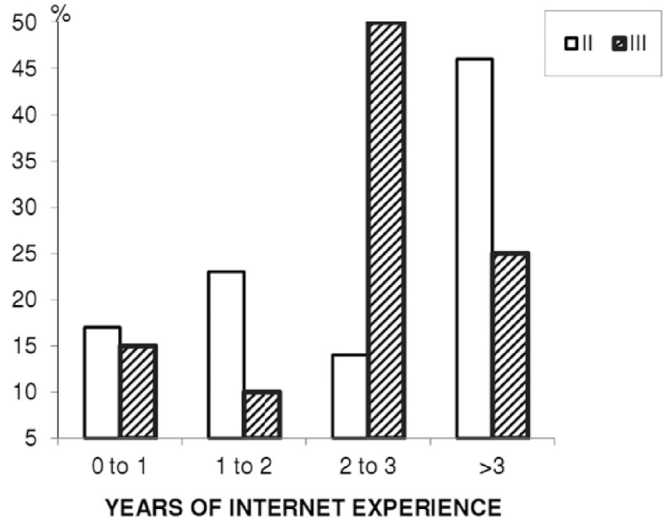

Fig. 1. Differences between second and third generation of students in internet experience.

of students.

Second year students spend significantly less time on FB (χ^2 (4) = 50.52, p < .0001, Cramer's V = .5026). Nearly half of the second year students (57%) versus 17% of the third year students spend on FB less than 30 min a day. 22% of the second year students spend between 30 min and 1 h on FB as compared to 40% of the third year respondents. None of the second year students spend more than 3 h, while among third year students there are those who spend more than 3 h a day on FB, which is shown in Fig. 2.

This fact is in accordance with the results of the analysis given in (Johnston, Tanner, Lalla, & Kawalski, 2013) which indicated that the time spent on Facebook per day felt within the 10–30 min category. Moreover, Christofides, Muise, and Desmarais (2009) showed that most users spend between 10 min and 1 h on Facebook daily.

The most common daily activity on the web for the participants is checking e-mail. 43% of the second year students check their mails – weekly (χ^2 (4) = 28.38, p < .0000, Cramer's V = .38).

Somewhat less than half of the students in the second year, and half of the third year students have more than 300 friends on FB. The higher percentage of second year students has between 100 and 200 FB friends. In the category where the number of friends is between 200 and 300, the ratio between the two generations is reversing, as illustrated in Fig. 3. Only a small percentage of the new generation students have more than 600 friends. The differences between two generations of students regarding the number of FB friends are slightly significant (χ^2 (3) = 15.66, p < .0013, Cramer's V = .28).

Second year students update their profiles more frequently than third year students (χ^2 (4) = 89.88, p < .000, Cramer's V = .67). The same stands for writing on their own (χ^2 (5) = 59.15, p < .0001, Cramer's V = .54), and their friends' walls (χ^2 (4) = 19.37, p < .0001, Cramer's V = .31). The majority of respondents rarely uses FB seeking new friends and generally, rarely participates in creating groups, as well as areas of interest (Table 1). Thereby, third year students are more frequently engaged in creating groups than their younger colleagues (χ^2 (4) = 35.69, p < .0001, Cramer's V = .42).

Membership in groups is more prominent in third year students (70% of them have joined up to 20–40 groups) (χ^2 (3) = 63.87, p < .000, Cramer's V = .57). The enrolment of second year students into groups shows a different pattern. They tend to be members of either a small or a large number of groups, as presented in Fig. 4. Both groups show almost no tendency to be either creators or administrators in their groups (Fig. 5).

The new generation of students was more excited when joining FB, in comparison to the other respondents (χ^2 (4) = 32.47, p < .0001, Cramer's V = .40), as shown in Fig. 6. Also, most of the students in both generations reported they are not addicted to FB. These attitudes are more pronounced in the younger group, whereas 25% of older students are willing to admit addiction (χ^2 (4) = 18.73, p < .0009, Cramer's V = .31), as shown in Fig. 7.

Approximately the same number of students from each group uses FB as a tool to find new friends. On the other side, almost half of the second year students deny that FB is used for this purpose (χ^2 (4) = 37.18, p < .0001, Cramer's V = .43). Most students from both groups use FB to stay in touch with friends from high school. This behaviour is much more expressed among the second year students (χ^2 (4) = 25.71, p < .0001, Cramer's V = .36). Second year students show the highest level of agreement with the statement which refers to knowledge of the activities of their friends, while in concern with the same statement, half of the third year students are neutral (χ^2 (4) = 44.82, p < .0001, Cramer's V = .47). Concerning the

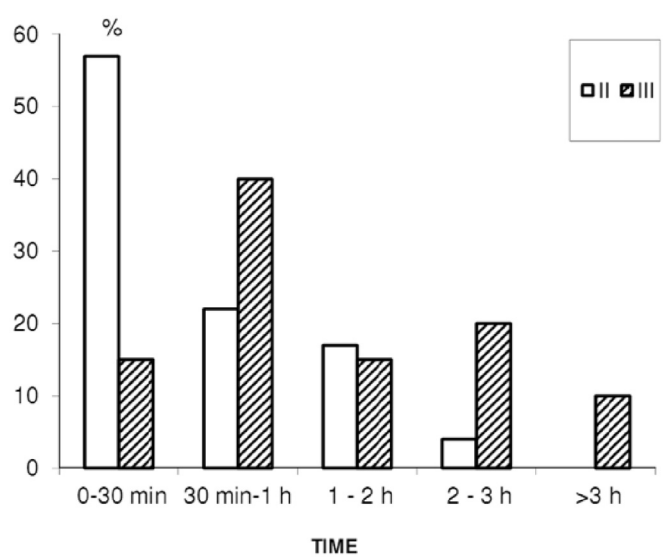

Fig. 2. Differences between two generations of students in daily internet usage.

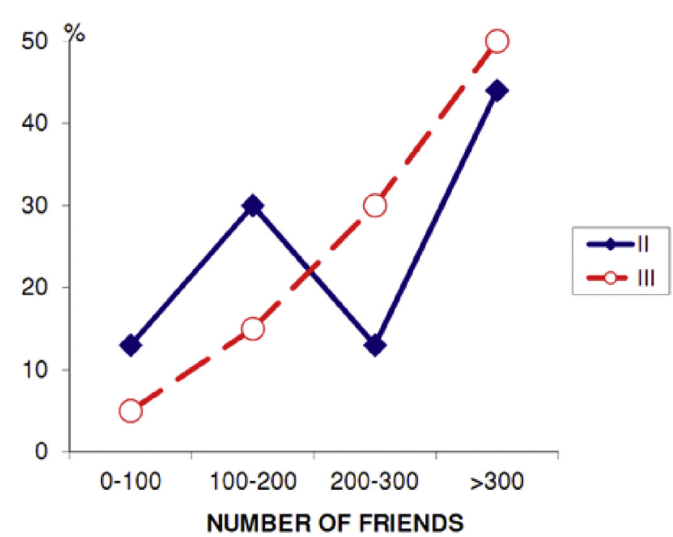

Fig. 3. Differences between two generations of students regarding to the number of FB friends.

Table 1
Frequency of Facebook wall use.

How often do you participate in the following activities on Facebook?	Hourly		Daily		Weekly		Monthly		Rarely		I don't do this	
	Two generations of students (%)											
	II	III	II	III	II	III	II	III	II	III	II	III
Update my profile	13	0	52	10	22	25	5	55	4	10	4	0
Write on my own wall	22	5	26	15	39	15	5	20	4	40	4	5
Write on my friend's wall	0	0	39	20	35	25	13	20	9	25	4	10
Search for new friends to add	9	5	4	10	9	15	30	20	35	35	13	15
Create groups	0	5	0	10	4	15	13	10	35	10	48	50
Create areas of interest	0	5	0	5	0	5	9	5	26	25	65	55

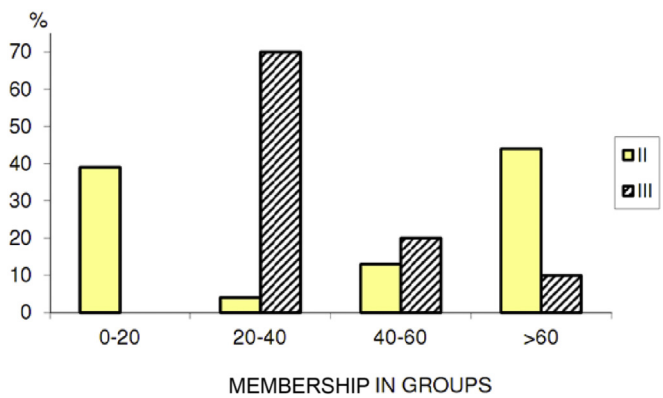

Fig. 4. Differences between two generations of students with respect to membership in groups.

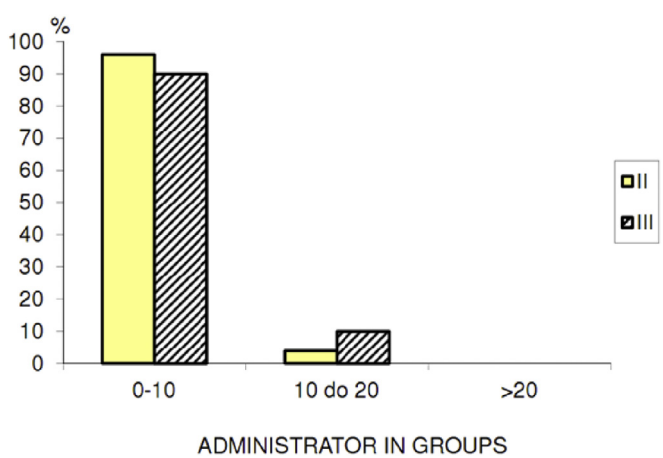

Fig. 5. Differences between two generations of students with respect to being group administrators.

connectedness and communication with FB friends, younger students showed mixed opinions, whereas a higher percentage of older students choose to be neutral (χ^2 (4) = 23.33, p < .0001, Cramer's V = .34). Over half of the respondents of both groups are more likely to believe that FB is not used to make social connections with people they never met in person. This viewpoint is emphasized among third year students (χ^2 (4) = 12.04, p < .017, Cramer's V = .25) (Table 2).

About half of both groups use the internet every day to find information, 40% of younger students were engaged in this activity less frequently (weekly), while older students surf the internet more frequently (χ^2 (4) = 32.55, p < .0000, Cramer's V = .40).

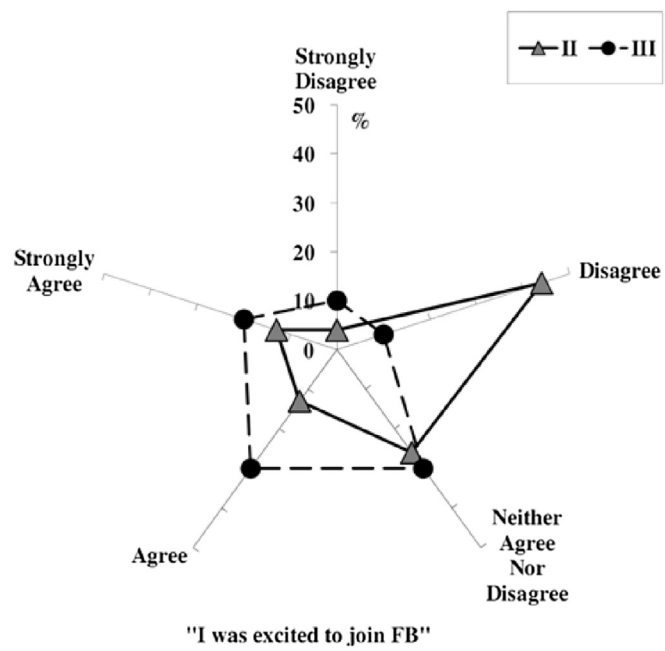

Fig. 6. Differences between two generations of students with regards to excitement when joining FB.

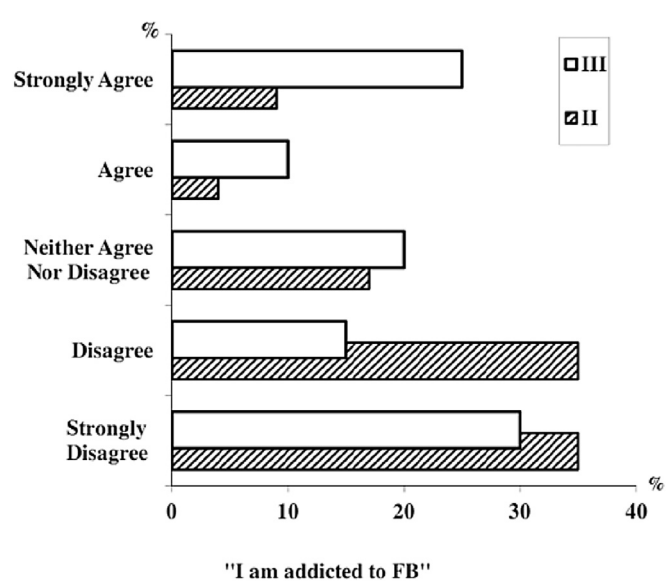

Fig. 7. Differences between two generations of students with regards to the willingness of admitting addiction to FB.

Table 2
Differences between two generations of students in attitudes towards connectedness on FB.

Being involved on Facebook	Strongly disagree		Disagree		Neither agree nor disagree		Agree		Strongly agree	
Two generations of students (%)	II	III	II	III	II	III	II	III	II	III
I use FB to make social connections that couldn't find in person	17	10	49	40	13	25	17	25	4	0
I use FB to keep track of what my friends are doing	4	10	26	25	30	50	40	4	0	10
I use FB to stay connected with friends from high school	0	0	9	5	9	35	69	40	13	20
I use FB to meet new people	4	10	52	20	9	10	35	40	0	20

Students show a similar pattern regarding the use of the internet for educational purposes, thereby the older generation is more inclined to use the internet for academic purposes (χ^2 (4) = 24.27, $p < .0001$, Cramer's V = .35) (Fig. 8).

Self-reported and objectively registered indicators, such as number or friends, hours per day or week college students spent on FB, frequency of wall posts, etc., reflect the behavioural pattern of FB users. The purpose of this paper is to outline the similarities and differences between two generations of college students concerning the FB usage. Table 3 provides a concise and transparent summary of the key research questions we are pretending to answer. The most important remarks, thus, could serve as a starting point for researchers in this field providing the opportunity to further explore generational differences in FB usage.

6. Discussion

Facebook appears to have lost its function as a mediator in creating new friendships, despite the fact that the time spent online as well as the number of FB users increase. Also, taking into account that the number of virtual friends with the purpose of information exchange rises, there is a slight awareness about the risk of addiction to FB. This can be explained by the longer usage of FB, or more willingness to attribute this behaviour to addiction. Finally, there is higher number of students who use FB for academic purposes.

Generally speaking, older social media users are more likely than younger to share certain types of information, as demonstrated by focus group discussions indicating that they have been waning enthusiasm for Facebook, disliking the increasing adult presence, but they keep using it because participation is an important part of overall teenage socializing (Madden et al., 2013).

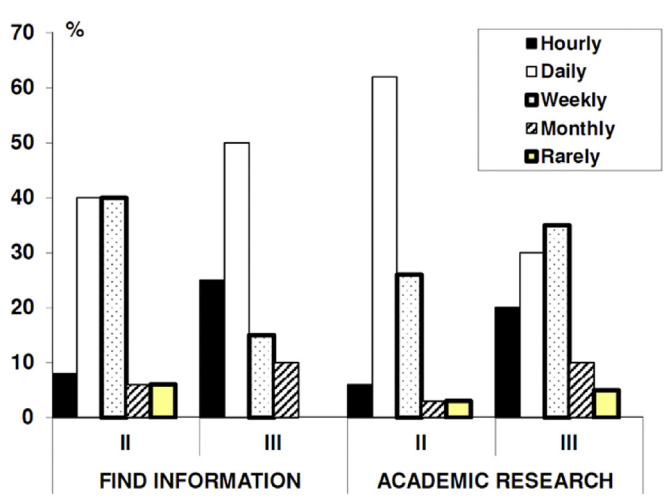

Fig. 8. Differences between two generations of students in FB usage for academic purposes.

The interaction between users is more motivated by the user who generated the content rather than by established friendship relations. So, new approaches should be emerging in social media using social currencies to promote interaction between users who are not explicitly friends. Kaltenbrunner, Gonzalez, De Querol, and Volkovich (2011) hypothesise that stimulating users to interact around content rather than only declaring friendships is very likely to generate a higher level of interaction in the network. As Johnston et al. (2013) argued, future research should make an attempt to solidify the outcomes of some of the research questions regarding more detailed and rigorous studies of demographic use of SNS, particularly FB, demographic breakdown of non-use of FB, as well as more formalised studies on the roles that age plays in the patterns and reasons of its usage.

The results have shown several significant differences between two generations of students. This is in accordance with the findings from the literature, with regards to the behaviour of students from various classes considering the usage of Facebook. So as, when students check up on friends, they are engaging in the practice of social information seeking which has been shown to be directly related to social capital (Ellison et al., 2011; Pascarella & Terenzini, 2005). When freshmen check up on friends, they are doing so in the context of needing to build new relationships, while students in other classes are more focused on maintaining their current relationships. Therefore, when freshmen students build social capital, they are building their social support network necessary to promote a sense of connection to the institution leading to a higher degree of academic commitment, which leads to improvement in academic performance. Perhaps seniors, throughout their time at university, have learnt appropriate self-regulation skills and apply these skills to their technology use.

Additionally, it is possible that seniors group's level of social capital has plateaued and they are less likely to use Facebook for relationship building and maintenance activities compared to their younger peers. The pattern of results suggests that activities involving interpersonal connections are more positively related to Grade Point Average (GPA), which is congruent with previous research showing that these types of activities are related to student engagement (Junco, 2012a; 2012b). These results may be explained by the demands faced by students at each level: freshmen must not only adapt to a new academic environment, but also a social one in order to be successful (Pascarella & Terenzini, 2005). Beyond their first year of college, students might use Facebook less for building friendships, which may have been reflected in how different types of usage were related to academic outcomes.

Furthermore, only multitasking with Facebook was significantly negatively predictive of GPA for sophomores and juniors but not for seniors. For freshmen, checking up on friends on Facebook was a stronger positive predictor of GPA than the time spent preparing for class. However, seniors spent significantly less time on Facebook, and doing their coursework, than students from other class ranks (Junco, 2015). Sophomores and juniors only exhibited a negative

Table 3
Characteristics and similarities between two generations of college students upon FB use.

Characteristics of the second year students	Characteristics of the third year students	Similarities between two college students generations
- lower number of FB users	- all students have FB account	- don't prefer to create groups
- longer experience in internet usage	- high percentage play online games	- the main motivation of FB usage is to stay connected with friends from high school
- better academic achievement	- checking e-mail more often	- not inclined to make social connections that couldn't find in person
- lower percentage play online games	- have more FB friends	- use internet every day to find necessary information and for academic research
- spend less time on FB	- higher rate of group membership, engagement in large number of different sorts of groups	
- update FB profile and write on their own FB wall more often	- more excited towards joining FB	
	- show more willingness to admit as being addicted to FB	

relation between multitasking on Facebook and GPA and seniors did not exhibit any relation between either of the Facebook use variables and GPA. Previous findings suggest there is a dynamic at play related to class standing that influences how students interact with Facebook and how such interactions are associated with academic outcomes. These results are congruent with research on the freshman experience, i.e. entering students have yet to learn important academic skills in order to be successful (Upcraft, Gardner, Barefoot, & Associates, 2005).

6.1. Limitations and further research

A number of areas for future studies within this field remain to be explored. This study is based on data gathered at one university, restricted to engineering students, and thus would benefit from being generalized by having data from several institutions. To be able to extrapolate further, similar researches might be conducted in different majors of students, analysing disciplinary variation comparing Facebook usage in a technical (engineering) and non-technical (social) disciplines, involving different backgrounds.

A future study would require a higher number of participants including students across all years. Collecting longitudinal data over a series of years, tracking incoming first year students through to graduation evaluating their technology use and social interactions each year, and following them after they graduate, is also a necessary next step, to evaluate how differences in Facebook use change over time.

There is a need for more qualitative studies of student and teacher perspectives on technology use for learning, from a broader range of universities. Qualitative data that could identify the complexities of student choice of technologies and in-depth patterns of use (context of social media use, not just the amount of time spent using them) by means of multiple and continuous measures of Facebook engagement as a supplement to the traditional survey methodology. Even though research shows that there is a high adoption rate of social technologies among college students, digital inequalities still persist (Junco, 2013).

Hence, while they are seldom taken into account, individual differences (i.e. personal characteristics of users, their socio-economic background, life circumstances, disciplinary differences, pedagogy and assessment regimes within which university students operate, the nature of technology use – what is the technology used for, as well as individual differences in terms of learning styles) among Facebook users have to be explored.

Analysing users of different social media platforms (e.g., Twitter, Instagram, etc.) separately to account for the different features they provide is also recommended. In addition, a few cross-country, transnational, cross-cultural and longitudinal studies that analyse the behavioural and academic outcomes of the continuous Facebook usage are recommended.

This study was not meant to be definitive in any sense, but rather a starting point in a promising direction of research that examines student behaviour during the course of study in order to gain better educational outcomes and prepare students for real life situations.

7. Conclusions

Based on the results of this study we can conclude that the number of users of FB grows among older students in comparison to younger one. Apart from that, second year students have more experience in internet use. There have been an increasing number of the third year students who are more prone to playing online games. The highest percentage of students in their second year spend 30 min a day on FB and no more than 3 h, while in the third year, the highest percentage of FB users spend between 30 min and 1 h. Furthermore, 10% of those participants spend more than 3 h on Facebook. These findings reflect the trends of the new millennial generation and their familiarity with online environments, i.e., the time spent online increases, as Facebook gains more users from the younger population of students.

There is also a tendency of increasing the number of friends in both generations, but it is more pronounced among third year students. Second year students update their profiles and write on the FB wall more often than their older counterparts. Both generations rarely use FB to make new friendships. Membership in groups is more popular within the third year students. Second year students deny that they were excited when joining FB. Most students in both groups use FB to stay in touch with friends from their high school. Also, most students do not consider that they are addicted to FB. This is particularly true for second year students, while 25% of the third year students are willing to admit addiction. When looking for information, about half of both generations surf the internet daily, the other half does it rarely, and among them, third year students search for academic contents more frequently.

The limitations of the study refer to the sample characteristics which included students from one university so may not be representative of the population of all Facebook users, and therefore, caution should be taken when generalizing the findings. Yet, this work provided an exploratory investigation of Facebook usage among university students with focus on gender differences. The greatest differences between the two generations of students were demonstrated for the following Facebook activities: frequency of updating of FB profile, writing on own FB wall, membership in groups and frequency of Facebook daily

usage, while the weakest, although statistically significant differences were recorded for the number of FB friends and creating various areas of interest.

We hope that we have managed to address and clarify some peculiarities related to generational differences in Facebook usage. The results of the study have revealed very interesting and significant generational differences in Facebook usage and related perceptions, which can then be the basis for future theoretical development and empirical validation. Furthermore, the study of FB usage should be extended in terms of different variables such as educational level, profession, cultural differences, etc., to enable the possible sources of these differences to be examined in an in-depth manner.

Acknowledgements

This work was partially supported and funded by the Ministry of Education, Science and Technological Development of the Republic of Serbia (projects no. 32025, 32048, 36022, 36006 and 179002).

References

Baek, K., Holton, A., Harp, D., & Yaschur, C. (2011). The links that bind: uncovering novel motivations for linking on Facebook. *Computers in Human Behavior, 27*(6), 2243–2248.

Blattner, G., & Fiori, M. (2009). Facebook in the language classroom: promises and possibilities. *Instructional Technology and Distance Learning, 6*, 17–28.

Boyd, D. M., & Ellison, N. B. (2007). Social network sites: definition, history, and scholarship. *Journal of Computer-Mediated Communication, 13*, 210–230.

Cattuto, C., & Hotho, A. (2011). Introduction to the special issue on social linking and hypermedia. *New Review of Hypermedia and Multimedia, 17*(3), 241–242.

Chang, C. W., & Heo, J. (2014). Visiting theories that predict college student's self-disclosure. *Computers in Human Behavior, 30*, 79–86.

Chen, Y. C. (2011). Learning styles and adopting Facebook technology. In *Proceedings of technology management in the energy smart world (PICMET 2011), 31 July – 4 August 2011* (pp. 1–9).

Christofides, E., Muise, A., & Desmarais, S. (2009). Information disclosure and control on Facebook: are they two sides of the same coin or two different processes? *Cyberpsychology & Behavior, 12*(3), 341–345.

Čičević, S., Nešić, M., & Samčović, A. (2012). Evaluation of students' attitudes towards online and blended learning. In *Proceedings of the 6th international conference on technology, education and development INTED 2012. Valencia, Spain, 5–7 March 2012* (pp. 3874–3881).

Čičević, S., Samčović, A., & Čubranić-Dobrodolac, M. (2012). Students' attitudes towards learning in educational environment. *International Journal of Computer Science Research and Application, 2*(1), 101–108.

Cotten, S. R. (2008). Students' technology use and the impacts on well-being. In R. Junco, & D. M. Timm (Eds.), *Using emerging technologies to enhance student engagement* (pp. 55–70). San Francisco, CA: Jossey-Bass. New directions for student services, Issue #124.

Dawson, S. (2008). A study of the relationship between student social networks and sense of community. *Educational Technology & Society, 11*(3), 224–238.

Dong, Q. (2008). Explaining why young adults use MySpace and Facebook through uses & gratification theory. In *Annual meeting of the NCA 94th annual convention, San Diego, California, USA, 20 November 2008*.

Drigas, A. S., Ioannidou, R. D., Kokkalia, G., & Lytras, M. (2014). ICTs, mobile learning and social media to enhance learning for attention difficulties. *Journal of Universal Computer Science, 20*(10), 1499–1510.

Drouina, M., O'Connor, K. W., Schmidt, G. B., & Miller, D. A. (2015). Facebook fired: legal perspectives and young adults' opinions on the use of social media in hiring and firing decisions. *Computers in Human Behavior, 46*, 123–128.

Ellison, N., Steinfield, C., & Lampe, C. (2007). The benefits of Facebook "friends": social capital and college students' use of online social network sites. *Journal of Computer Mediated Communication, 12*(4), 1143–1168.

Ellison, N., Steinfield, C., & Lampe, C. (2011). Connection strategies: social capital implications of Facebook-enabled communication practices. *New Media & Society, 13*(6), 873–892.

Facebookcom. (2014). Retreived from: http://www.statisticbrain.com/facebook-statistics.

Flynn, N. (2009). *Facebook, take 2: Cyberbullying. Education week*. Retreived from http://blogs.edweek.org/edweek/LeaderTalk/2009/05/facebook_take_2_cyberbullying.html?qs=facebook.

Gray, R., Vitak, J., Easton, E. W., & Ellison, N. B. (2013). Examining social adjustment to college in the age of social media: factors influencing successful transitions and persistence. *Computers and Education, 67*, 193–207.

Hayes, M., van Stolk-Cooke, K., & Muench, F. (2015). Understanding Facebook use and the psychological affects of use across generations. *Computers in Human Behavior, 49*, 507–511.

Johnston, K., Tanner, M., Lalla, N., & Kawalski, D. (2013). Social capital: the benefit of Facebook 'friends'. *Behaviour & Information Technology, 32*(1), 24–36.

Jones, S., & Fox, S. (2009). *Generations online in 2009. Pew internet and American life project*. Retrieved from http://www.pewinternet.org/w/media/Files/Reports/2009/PIP_Generations_2009.pdf.

Junco, R. (2012a). The relationship between frequency of Facebook use, participation in Facebook activities, and student engagement. *Computers & Education, 58*, 162–171.

Junco, R. (2012b). In-class multitasking and academic performance. *Computers in Human Behavior, 28*(6), 2236–2243.

Junco, R. (2013). Inequalities in Facebook use. *Computers in Human Behavior, 29*(6), 2328–2336.

Junco, R. (2015). Student class standing, Facebook use, and academic performance. *Journal of Applied Developmental Psychology*, 18–29.

Kaltenbrunner, A., Gonzalez, G., De Querol, R., & Volkovich, Y. (2011). Comparative analysis of articulated and behavioural social networks in a social news sharing website. *New Review of Hypermedia and Multimedia, 17*(3), 243–266.

Karlin, S. (2007). Examining how youths interact online. *School Board News, 73*, 6–9.

Kim, D., Hwang, E., & Choi, H. G. (2013). A user opinion and metadata mining scheme for predicting box office performance of movies in the social network environment. *New Review of Hypermedia and Multimedia, 19*(3–4), 259–272.

Ku, H. Y., Tseng, H. W., & Akarasriworn, C. (2013). Collaboration factors, teamwork satisfaction, and student attitudes toward online collaborative learning. *Computers in Human Behavior, 29*(3), 922–929.

Lenhart, A. (2008). *Writing, technology, and teens. Pew internet and American life project*. Retrieved from http://www.pewinternet.org/w/media//Files/Reports/2008/PIP_Writing_Report_FINAL3.pdf.

Madden, M., et al. (2013). *Teens and technology 2013*. Pew Research Center. Retrieved from http://www.pewinternet.org/Reports/2013/Teens-and-Tech.aspx.

Manca, S., & Ranieri, M. (2013). Is it a tool suitable for learning? A critical review of the literature on Facebook as a technology – enhanced learning environment. *Journal of Computer Assisted Learning, 29*(6), 487–504.

Marketingmreza.rs. (2014). Retreived from: http://marketingmreza.rs/tag/iab-srbija.

Mazer, J. P., et al. (2007). I'll see you on "Facebook": the effects of computer-mediated teacher self-disclosure on student motivation, affective learning, and classroom climate. *Communication Education, 56*, 1–17.

Mazman, S. G., & Usluel, Y. K. (2011). Gender differences in using social networks. *TOJET: The Turkish Online Journal of Educational Technology, 10*(2), 133–139.

Nagel, L., & Kotze, T. (2010). Supersizing e-learning: what a CoI survey reveals about teaching presence in a large online class. *The Internet and Higher Education, 13*, 45–51.

Park, N., Lee, S., & Kim, J. H. (2012). Individuals' personal network characteristics and patterns of Facebook use: a social network approach. *Computers in Human Behavior, 28*, 1700–1707.

Pascarella, E. T., & Terenzini, P. T. (2005). *How college affects students*. San Francisco, CA: Jossey-Bass.

Pasek, J., & Hargittai, E. (2009). Facebook and academic performance: reconciling a media sensation with data. *First Monday, 14*(5).

Pejčić-Tarle, S., Čičević, S., Davidović, M., Petrović, D., & Bojković, N. (2007). Introducing elearning concept at the faculty of transport and traffic engineering, university of Belgrade – small steps towards big achievements. In *Proceedings of the elearning conference: elearning toward effective, education and training in the information society. WUS Austria, project report summary* (pp. 74–80).

Peters, D., Gareth, J., & Peters, D. (2005). *Learning styles and approaches to studying in sports-related programmes: Relationships to academic achievement and implications for successful learning, teaching and assessment* (pp. 1–6). University College Worcester. Project Report Summary, LB2300 Higher Education.

Prensky, M. (2001). Digital natives, digital immigrants: do they really think differently? *On the Horizon, 9*(6), 1–6.

Quan-Haase, A. (2007). College students' local and distant communication: blending online and offline media. *Information, Communication and Society, 10*, 671–693.

Roberts, D. F., & Foehr, U. G. (2008). Trends in media use. *The Future of Children, 18*, 11–37.

Roblyer, M. D., et al. (2010). Findings on Facebook in higher education: a comparison of college faculty and student uses and perceptions of social networking sites. *Internet and Higher Education, 13*, 134–140.

Salaway, G., et al. (2007). *The ECAR study of undergraduate students and information technology*. Boulder, Colorado: EDUCAUSE 2007. Retrieved from http://connect.educause.edu/library/abstract/TheECARStudyof Underg/45075.

Samčović, A., & Čičević, S. (2013). Security aspects of the social network Facebook: some empirical results. In *Proceedings of the international conference on applied internet and information technologies ICAIIT 2013, Zrenjanin, Serbia, 25 October 2013* (pp. 244–247).

Shailey, S. (2009). A case study-based investigation of students' experiences with social software tools. *New Review of Hypermedia and Multimedia, 15*(3), 245–265.

Teng, C. I. (2015). Drivers of interdependence and network convergence in social networks in virtual communities. *Electronic Commerce Research and Applications, 14*(3), 204–212.

Upcraft, M. L., Gardner, J. N., Barefoot, B. O., & Associates (Eds.). (2005). *Meeting challenges and building support: Creating a climate for first-year student success*. San Francisco, CA: Jossey-Bass.

Webrzs.stat.gov.rs. (2014). Retreived from: http://webrzs.stat.gov.rs/WebSite.
Xi, Z., Liu, L., & Ordóñez de Pablos, P. (2014). The auxiliary role of information technology in teaching: enhancing programming course using Alice. *International Journal of Engineering Education, 30*(3), 560–565.
Young, J. R. (2009). How not to lose face on Facebook, for professors. *The Chronicle of Higher Education, 55*, A1–A13.
Zhang, X., Ordóñez de Pablos, P., & Zhang, Y. (2012). The relationship between incentives, explicit and tacit knowledge contribution in online engineering education project. *International Journal of Engineering Education, 28*(6), 1341–1346.